Google Play Store App Market

EXPLORATORY DATA ANALYSIS & MARKET SEGMENTATION REPORT

1. Project Background & Scope

The Android mobile app ecosystem represents a highly dynamic marketplace driven by multi-faceted user preferences, complex pricing models, and distinct engineering optimization requirements. This report delivers an empirical review of the Google Play Store landscape based on the structural evaluation of over 10,000 applications.

The primary objective of this Exploratory Data Analysis (EDA) is to parse, clean, and map application metadata (including ratings, size metrics, install bands, and monetization mechanisms) to isolate structural patterns that drive store listings and marketplace performance.

10,841	92.6%	1,474
RAW APP ENTRIES	FREE APPLICATION SHARE	MISSING RATINGS RESOLVED

2. Data Engineering & Cleansing Pipeline

Before conducting market segmentations, raw dataset attributes were passed through a robust data cleaning pipeline via a structured Jupyter notebook (`gplay_prject.ipynb`). This process addressed missing metrics, unformatted text strings, and inconsistent units:

- **Structural Cleanup:** Extraneous variables such as `Unnamed: 0` were dropped from memory to preserve memory allocation and maintain database integrity.
- **Rating Imputation:** The `Rating` variable presented 1,474 missing values. These were systematically handled by executing mean-imputation to preserve data distribution without altering standard metrics.
- **Unified Size Mapping (`cleansize` function):** The raw `Size` attribute contained mixed string labels (e.g., Megabytes 'M' and Kilobytes 'k'). A custom mapping pipeline converted all objects to a consistent floating-point value tracking Megabytes (MB). Entries labeled as "Varies with device" were filled using calculated statistical column averages.
- **Numeric Type Casting:** Special string characters, currency indicators (`$`), plus signs (`+`), and comma thousands-separators (`,`) were removed from the `Installs` and `Price` arrays to safely cast them into `int64` and `float64` fields.

Data Quality Note

Rows containing isolated null observations in non-numeric deterministic categorical fields like **Type** and **Content Rating** were excluded from downstream evaluations, preserving 10,840 complete rows.

3. Market Structure & Monetization Models

The monetization topology of the store leans heavily toward no-upfront-cost structures. Out of the evaluated listing records, **10,039 apps are Free**, compared to just **800 Paid apps**. This sets the market base at roughly ~92.6% free software distribution, proving that in-app advertising or freemium multi-tiered subscriptions represent the primary business models on the platform.

Top 5 App Categories by Average Price

When exploring premium software pricing strategies across various business verticals, certain categories command significantly higher listing price-points. The table below outlines the premium pricing leaders:

App Category	Average Premium Listing Price	Primary Target Audience Segment
FINANCE	$7.93	Professionals, Day-traders, High-Net-Worth Individuals
LIFESTYLE	$6.18	Boutique Services, Niche Concierge Organizers
MEDICAL	$3.11	Healthcare Operators, Diagnostic Tools, Practitioners
EVENTS	$1.72	Corporate Event Organizers, Booking Agents
FAMILY	$1.23	Parents, Children Education Modules, Safety Frameworks

4. Key Analytical Takeaways

- **Finance and Lifestyle Dominance:** High listing averages in Finance and Lifestyle point to the presence of niche, high-priced specialty utility tools or premium vanity apps.
- **Storage Footprint Optimization:** Standardizing software size into unified Megabytes (MB) allows development engineering squads to build predictive cross-correlation tracks between local file storage space and consumer retention levels.
- **Consumer Engagement:** High install counts are strongly correlated with free access points, shifting the data engine's attention toward user-retention and volume-based monetization models rather than simple upfront paid conversions.